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PHD PRESENTATION SAMPLE

Prepared by:

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SOLUTIONS PVT. LTD.
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+91-8093778526



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Cancer Defence Deck Sample Preview

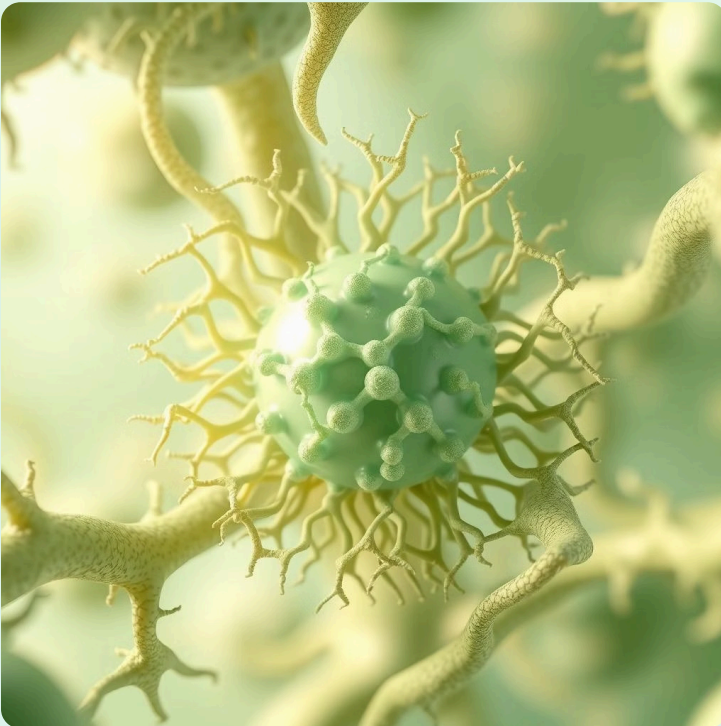
CANCER BIOLOGY

MINIMAL

PPTX SAMPLE

Adapted from an open-access UCL PhD thesis on 3D in vitro modelling of cancer invasion.

Objective of this sample: ***demonstrate clean storyline, typography consistency, and visual hierarchy in a 7-slide defence format.***



3D In Vitro Model Workflow



Tumoroid setup



AFM + assays

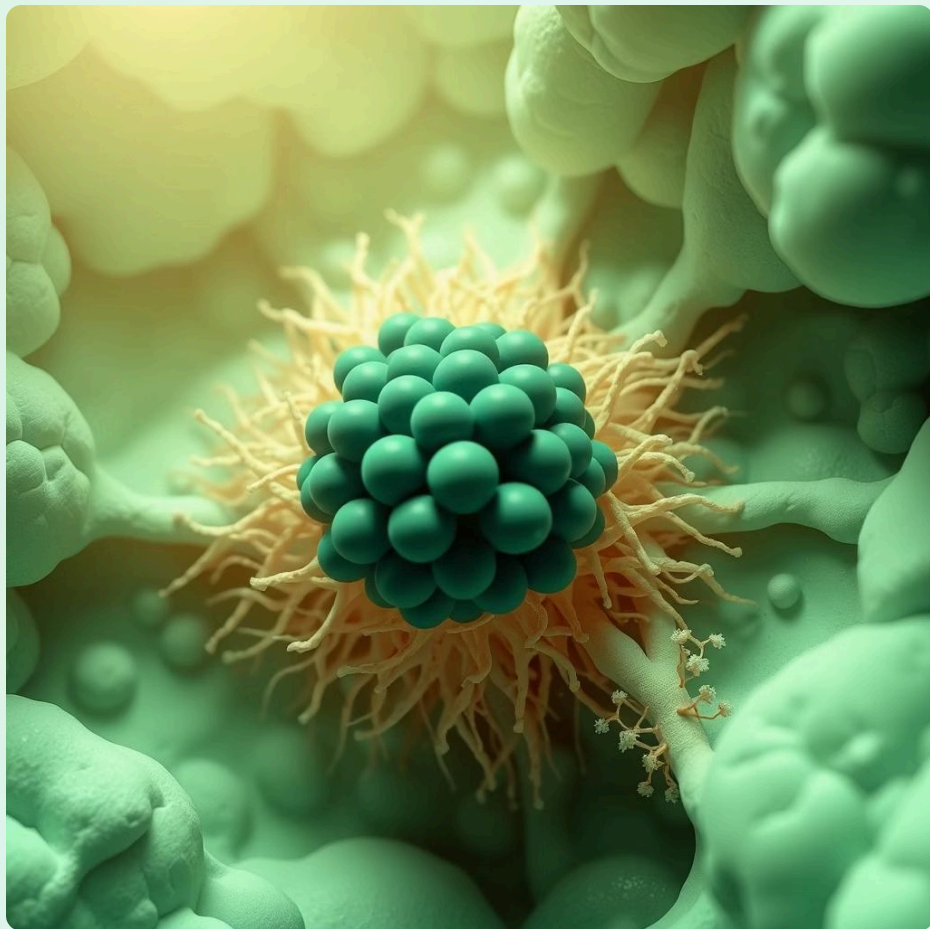


Invasion readout

Adapted from Micalet, "The Bio-Mechanical Signatures of Cancer Invasion Using a 3D In Vitro Model" (UCL, 2024)

Why this problem matters

Cancer invasion is shaped by the tumour microenvironment—not just by tumour cells themselves.



Traditional 2D models often fail to capture the critical mechanical signaling, matrix architecture, and stromal interactions that drive cancer invasion. A biomimetic 3D approach is essential to understand the physical microenvironment's role in invasive behavior.

Adapted from Micalet, "The Bio-Mechanical Signatures of Cancer Invasion Using a 3D In Vitro Model" (UCL, 2024)

Conceptual view

Tumour core

Central mass of tumour cells

Invasion

Tumour cells spreading into surrounding tissue

Matrix stiffness changes

Altered extracellular matrix mechanics

Cancer-associated stromal signals

Signals from surrounding stromal cells

Research gap and question

Understanding the research gap is crucial for defining the precise questions this thesis aims to answer, moving from existing knowledge to novel insights.



Literature Review

Synthesize existing studies



Identify Limitations

Highlight weaknesses of 2D models



Research Gap

Need for a 3D biomimetic approach



Research Question

Define precise investigational aim

What is already known

- Stiffness and mechanical signalling play a major role in tumour progression.
- Cancer cells can remodel their surrounding matrix directly and indirectly through stromal cells.
- Matrix-remodelling pathways such as MMPs and uPA are linked to invasion.

What this thesis addresses

- How do cancer cells alter matrix stiffness in a biomimetic 3D setting?
- Do invasive and less invasive phenotypes remodel the matrix differently?
- Can targeted perturbation of remodelling pathways reduce invasion?
- How do patient-derived fibroblasts relate to matrix remodelling patterns?

 **Core question: can a 3D in vitro model reveal measurable biomechanical signatures of cancer invasion—and show which pathways are actionable?**

Adapted from Micalet, "The Bio-Mechanical Signatures of Cancer Invasion Using a 3D In Vitro Model" (UCL, 2024)

Aim and study design

Aim: build a 3D tumour model that links microenvironment mechanics to cancer invasion and tests whether remodelling pathways can be suppressed.



Model

Engineer tumoroids in matrices with biologically relevant stiffness



Measure

Quantify stiffness change with AFM indentation tests



Profile

Assess matrix-remodelling genes and proteins



Perturb

Use inhibitors or CRISPR-Cas9 against key remodelling pathways



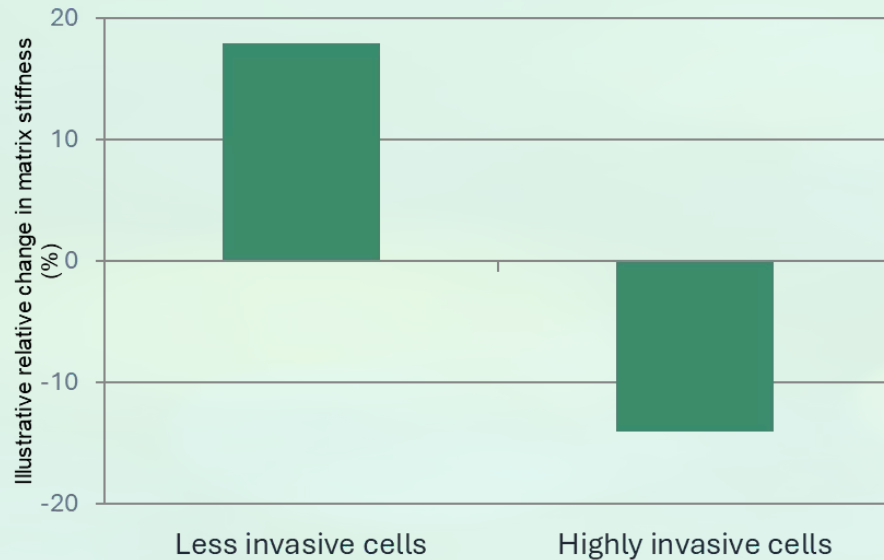
Translate

Incorporate patient-derived CAFs and compare remodelling patterns

Adapted from Micalet, "The Bio-Mechanical Signatures of Cancer Invasion Using a 3D In Vitro Model" (UCL, 2024)

Key finding 1: invasive phenotype changes the matrix differently

In the 3D model, less invasive cells tended to stiffen the microenvironment, while more aggressive cells softened it.

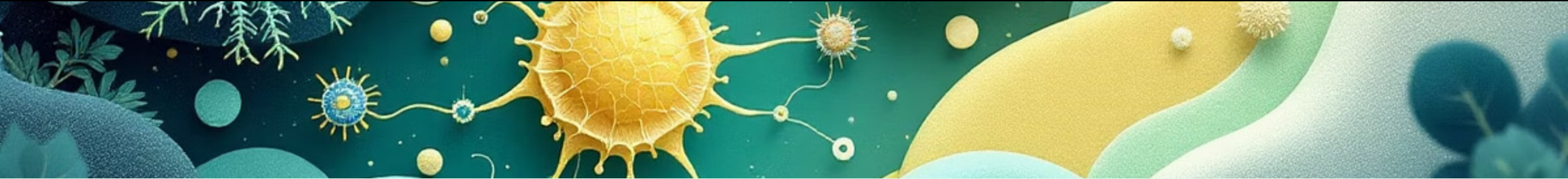


Takeaway

Biomechanical response is phenotype-dependent, not uniform.

- Changes in stiffness were linked to both active cell-generated forces and enzymatic ECM remodelling.
- This supports the use of mechanics as a measurable readout of invasive potential.

Adapted from Micalet, "The Bio-Mechanical Signatures of Cancer Invasion Using a 3D In Vitro Model" (UCL, 2024)

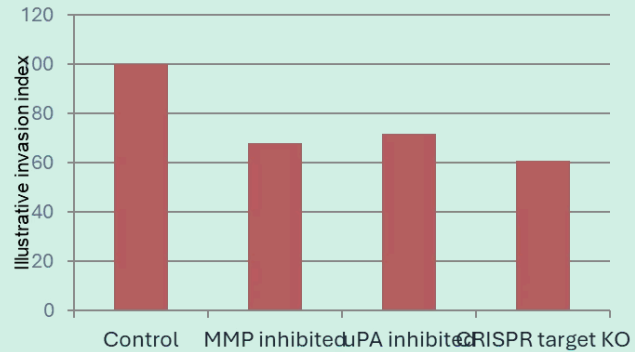


SAMPLE PREVIEW · ADAPTED FROM OPEN-ACCESS THESIS

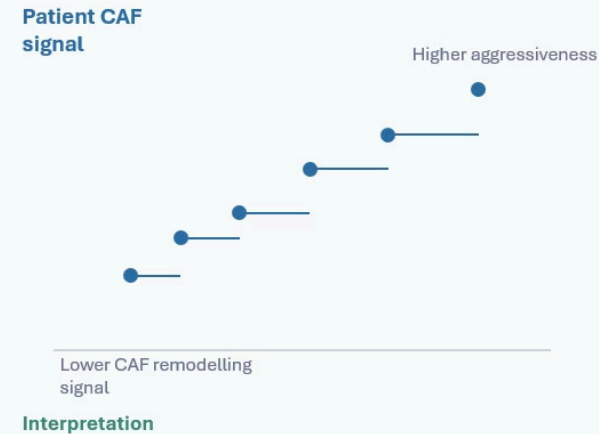
Key finding 2: remodelling pathways are actionable

Targeting matrix-remodelling activity reduced invasion, while patient-derived CAF behaviour tracked with disease aggressiveness.

Pathway intervention



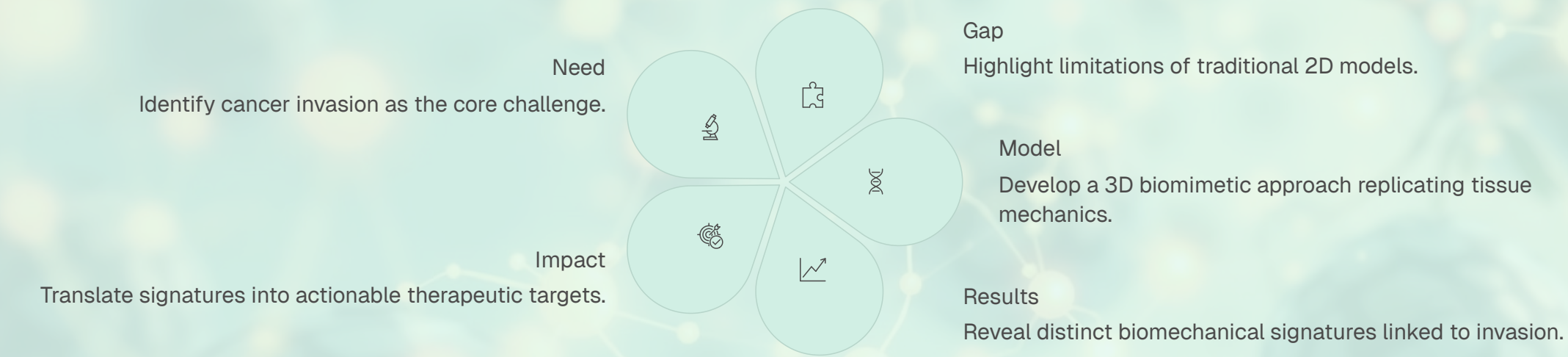
Patient CAF signal



☐ **Interpretation:** The model is useful not only for observation, but also for testing intervention points and stratified stromal behaviour.

Adapted from Micalet, "The Bio-Mechanical Signatures of Cancer Invasion Using a 3D In Vitro Model" (UCL, 2024)

Conclusion and storyline



This diagram illustrates the journey of the thesis, from identifying the fundamental challenge of cancer invasion to proposing a novel 3D model, discovering biomechanical signatures, and ultimately identifying actionable therapeutic targets.

		
Biomimetic 3D Model Established a new in vitro system to study invasion mechanically.	Quantifying Remodelling Showed matrix patterns differ by invasive phenotype and are measurable.	Actionable Targets Demonstrated remodelling pathways offer future therapy design targets.

Thank you

Adapted from Micalet, "The Bio-Mechanical Signatures of Cancer Invasion Using a 3D In Vitro Model" (UCL, 2024)

THANK YOU

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